# USRP X310 Data Recording Report

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**Abstract**

This report describes the data recording process using the USRP X310 device. This reciver is placed in a fixed position in São José dos Campos, SP, Brazil, in the GPS/GNSS Laboratory at Aeronautics Institute of Technology (ITA)[1]. The present report is updated as new recordings are made and new information is available.




Figure 1: Location of the GNSS antenna.

[1]https://github.com/ita-gnss-lab

# Contents


**I Recording overview** **3**

**1 Methodology** **3**

**2 Recording calendar** **4**

**3 Validating the recordings with GNSS-SDR** **4**
3.1 Signal source . . . . . 5
3.2 Signal conditioner . . . . . 6
3.3 Acquisition . . . . . 6
3.4 Tracking . . . . . 6
3.5 Telemetry decoder . . . . . 7
3.6 Observables . . . . . 7
3.7 PVT . . . . . 7

**II Recording reports** **8**

**4 Records on 08/10/2026** **8**
4.1 First record . . . . . 8

**5 Records on 13/10/2026** **9**
5.1 First record . . . . . 9
5.2 Second record . . . . . 10

**6 Records on 22/10/2026** **11**
6.1 First record . . . . . 11
6.1.1 Signal source . . . . . 11
6.1.2 Signal conditioner . . . . . 13
6.1.3 Acquisition . . . . . 13
6.1.4 Tracking . . . . . 15
6.1.5 Telemetry decoder . . . . . 16
6.1.6 Observables . . . . . 17
6.1.7 PVT . . . . . 18

# Part I
# Recording overview

This part introduces an overview os the recordings presented in this report. It includes the methodology used to record the data, the calendar of the recorded days, the configuration of the USRP X310, and the processing of the recordings with GNSS-SDR.

## 1 Methodology

The recording is computed automatically by a Linux service which is driven by the following guidelines:

- Everyday, the program will start within the time rage [4PM, 6PM]
- It will sense whether the USRP is detectable via `uhd_find_devices --args=''addr=192.168.10.2''`.
- If not found, it will exit silently and will sense again in intervals of 15 min.
- Once found, two recordings are made, each with duration of 5 minutes (300 s). The first recording happens right after the USRP is detected, and the second recording is made 30 min after the first one. After that, no more recordings will be made for that day.
- The two recordings of a day use different signal pairs:
  - First recording: channel 0 (RX0) GPS L1 C/A (1575.42 MHz), 4 MS/s; channel 1 (RX1) GPS L5 (1176.45 MHz), 20 MS/s; both 25 dB of gain, signed 8-bit I/Q (2 bytes per complex sample);
  - Second recording: channel 0 (RX0) GPS L1 C/A (1575.42 MHz); channel 1 (RX1) GPS L2C (1227.60 MHz); both 4 MS/s, 25 dB of gain, signed 16-bit I/Q (4 bytes per complex sample).
- For each recording a SigMF archive file is created, with the interleaved I/Q samples + metadata, like number of channels (dual channel recording), frequencies, sampling rate, gain, sample type, etc. Each channel has its own configuration, so the metadata of each channel is reported separately.
- Each recording is processed by the GNSS-SDR software receiver right after it is made, and the outputs of every receiver stage are added under the recording in Section II (see Section 3).
- After 15 days of records from the last report. Rubem will send an automatic message to header of the Laboratory, Prof. Dr. Felix Antreich, via WhatsApp with the summarized information from all recorded metadata;

## 2 Recording calendar

Table 1 shows the days when recordings were made. The color of each day indicates how many recordings were made on that day: green means 2 recordings, blue means 1 recording, and no color means no recording was made.

October

| M | T | W | T | F | S | S |
|---|---|---|---|---|---|---|
| | | | 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 12 | 13 | 14 | 15 | 16 | 17 | 18 |
| 19 | 20 | 21 | 22 | 23 | 24 | 25 |
| 26 | 27 | 28 | 29 | 30 | 31 | |

Table 1: Recording calendar.

## 3 Validating the recordings with GNSS-SDR

A recording is only useful if a GNSS receiver can use it: find the satellites in it, track their signals, decode their navigation messages, and compute a position. To show that, every recording is processed with GNSS-SDR[2], an open-source software-defined GNSS receiver, and the output of each of its stages is added to the report of that recording (Section II). Figure 2 shows the data chain of GNSS-SDR. Each channel (one per satellite signal) runs its own acquisition, tracking and telemetry decoder; observables and PVT combine all of them.

[2] https://gnss-sdr.org, v0.0.20.

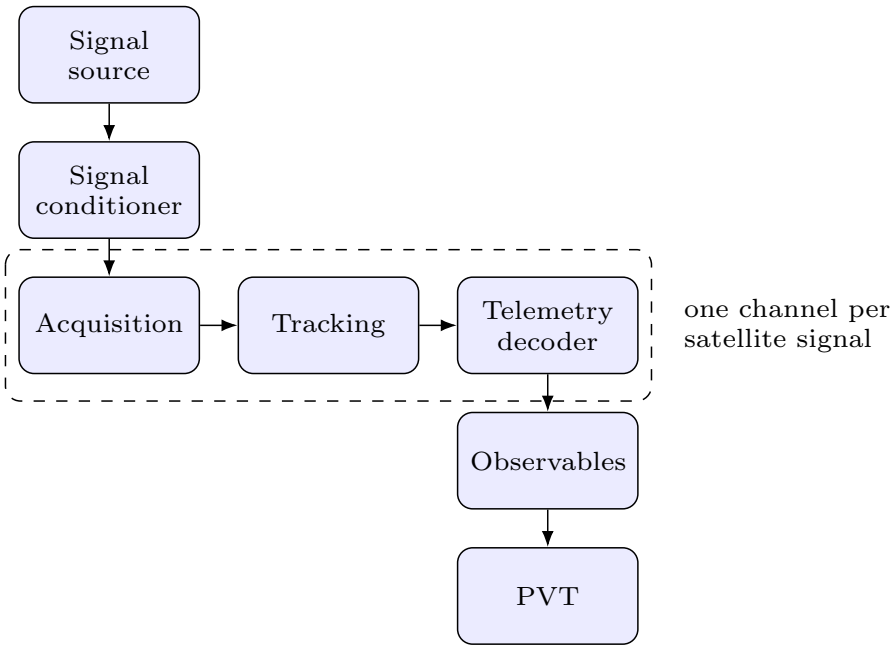



Figure 2: GNSS-SDR receiver chain.

**Processing setup.** Both channels of a recording are read by one `Multichannel_File_Signal_Source`, so the two bands advance in lock-step (with one independent source per band, one band runs ahead of the other and its measurements are discarded). That source reads both files at one rate: the 4 MS/s L1 C/A file of the L1/L5 recordings is upsampled 5 times (each sample repeated) to the 20 MS/s of L5 before GNSS-SDR. GNSS-SDR v0.0.20 computes its own position from L1 C/A only (broadcast ionosphere model): its dual-frequency solver ignores L2C altogether and was unstable with L5 on these recordings. The ionosphere-free, dual-frequency positioning is therefore computed outside GNSS-SDR's output, but using its observables and decoded ephemeris to generate our own single-point positioning (`gnss_spp.py`).

## 3.1 Signal source

Reads the recorded I/Q samples (here, the SigMF data files) and delivers them as a stream, one per channel. GNSS-SDR has no output of its own here, so the recording itself is examined.

- **Power spectral density** per channel: the shape of the receiver passband over a flat thermal-noise floor. GNSS signals are 15–20 dB below the noise, so they are invisible here; a narrow peak would be interference.
- **I/Q histograms**: should be Gaussian, centred on zero, and far from the integer limits (no clipping).

- **Table**: format, rate and gain per channel, I/Q standard deviation and mean, percentage of clipped values, and samples lost during the recording (overflows, zero-filled).

## 3.2 Signal conditioner

Adapts the samples to what the channels process: converts the file format (signed 8- or 16-bit interleaved I/Q) to complex floating point (data type adapter), optionally filters (input filter) and resamples (resampler).

- **Table** per channel: file format, data type adapter, filter, resampling, and rate before and after.

## 3.3 Acquisition

Detects which satellites are present and gives a first, coarse estimate of each one's code delay and Doppler shift, by correlating the signal with the satellite's spreading code over a two-dimensional grid of delays and Doppler frequencies, and comparing the peak with a detection threshold (set by the probability of false alarm). With dual-frequency assistance, the second band (L5 or L2C) is only searched for satellites already tracked on L1, around their L1 Doppler scaled to the second frequency.

- **Delay–Doppler map** (surface plot) of the strongest detection per band: a single sharp peak well above the floor.
- **Table**: search attempts, detections and detected satellites, and the best test statistic relative to the threshold. Attempts on satellites that are not in view, or that do not broadcast that band (L5 is only on Block IIF and III satellites, L2C on IIR-M, IIF and III), are rejected.

## 3.4 Tracking

Follows each detected signal with a delay-lock loop (code) and a phase-lock loop (carrier), producing every 1–20 ms the prompt correlator output (which carries the navigation data bits), the carrier-to-noise density ratio $C/N_0$, the Doppler, the code, and carrier phases.

- $C/N_0$ **and Doppler versus time** per satellite and band: $C/N_0$ of roughly 35–50 dB-Hz for an open-sky antenna, and smooth Doppler curves following the satellite motion.
- **Prompt I/Q scatter** of the strongest satellite per band: with the carrier phase locked, the energy sits on the in-phase axis in two clusters (the ±1 data bits); without phase lock it would be a ring. Each point is one prompt correlator output, the received samples $x[n]$ multiplied by the local code replica $c$ aligned at the estimated code delay $\hat{\tau}$ and by the local carrier replica (estimated Doppler $\hat{f}_D$ and phase $\hat{\phi}$), summed over the $N$ samples of one integration period ($T_s$: sampling period):

$$P = \sum_{n=1}^{N} x[n]\, c[n-\hat{\tau}]\, e^{-j(2\pi \hat{f}_D n T_s + \hat{\phi})} = I + jQ. \tag{1}$$

- **Table** per satellite: median $C/N_0$, percentage of time phase-locked, losses of lock and time tracked.

## 3.5 Telemetry decoder

Synchronizes to the navigation-message bits and frames, checks their parity/CRC (Cyclic Redundancy Check), decodes the ephemeris (orbit and satellite clock), and time-stamps every measurement with the satellite's transmit time (time of week). L1 C/A carries the legacy LNAV message (6-s subframes); L2C and L5 carry CNAV (12-s and 6-s messages).

- **Timeline** per band and satellite of when a valid transmit time was available.
- **Table**: messages decoded, ephemeris obtained, and when the first valid time of week came.

## 3.6 Observables

Aligns all channels on common receiver epochs and forms, for each satellite and band, the pseudorange (from transmit and receive times), the accumulated carrier phase and the Doppler. These are the measurements a positioning algorithm uses; GNSS-SDR also writes them to a RINEX observation file.

- **Availability** of valid pseudoranges per band and satellite over the recording.
- **Second band minus L1 pseudorange** per satellite: the same constant for every satellite (the receiver's inter-band hardware bias) plus the slowly varying, satellite-dependent ionospheric delay difference (metres). It proves that both bands measure the same satellites consistently.
- **Table** per satellite: common epochs, mean, standard deviation and drift of that difference.

## 3.7 PVT

Computes position, velocity and receiver clock (PVT) from the pseudoranges and the decoded ephemeris. The antenna position was surveyed: −23.209433°, −45.874875°, 596.42 m (WGS-84), so the errors below are true errors, not just scatter.

- **Horizontal error scatter and east/north/up error versus time** for GNSS-SDR's L1 C/A solution and for the dual-frequency ionosphere-free solution. Single-frequency single-point positioning is expected to stay within a few metres horizontally and ~10 m vertically; the iono-free combination removes the ionospheric bias but amplifies the code noise about three times.
- **Sky plot** of the satellites used.
- **Table**: mean error (accuracy), standard deviation (precision), 2-D and 3-D RMS, and 95% horizontal error.

# Part II
# Recording reports

## 4 Records on 08/10/2026

### 4.1 First record

| Parameter | Channel 0 | Channel 1 |
|---|---|---|
| Start time | 2026-10-08 22:26:37 UTC | |
| Frequency | 1575.420 MHz (L1) | 1176.450 MHz (L5) |
| Duration | 300.0 s | |
| Sample rate | 2.000 MS/s | |
| Sample type | signed complex 16-bit samples (I+Q), little-endian | |
| Gain | 20.0 dB | 20.0 dB |
| Antenna | RX1 | RX2 |
| File size | 2400.0 MB | 2400.0 MB |
| Samples written | 600,000,000 | 600,000,000 |
| Lost samples | 0 | 0 |
| Clock source | external | |
| Time source | internal | |
| Motherboard | USRP X310 | |
| Daughterboard | TwinRX RX0 | TwinRX RX1 |
| Recorder | UHD Python API | |

Table 2: Metadata for recording `x310_20261008T222630Z`, captured starting 2026-10-08T22:26:37.215640Z.

# 5 Records on 13/10/2026

## 5.1 First record

| **Parameter** | **Channel 0** | **Channel 1** |
|---|---|---|
| Start time | 2026-10-13 21:57:56 UTC | |
| Frequency | 1575.420 MHz (L1) | 1176.450 MHz (L5) |
| Duration | 300.0 s | |
| Sample rate | 2.000 MS/s | |
| Sample type | signed complex 16-bit samples (I+Q), little-endian | |
| Gain | 20.0 dB | 20.0 dB |
| Antenna | RX1 | RX2 |
| File size | 2400.0 MB | 2400.0 MB |
| Samples written | 600,000,000 | 600,000,000 |
| Lost samples | 0 | 0 |
| Clock source | external | |
| Time source | internal | |
| Motherboard | USRP X310 | |
| Daughterboard | TwinRX RX0 | TwinRX RX1 |
| Recorder | UHD Python API | |

Table 3: Metadata for recording `x310_20261013T215749Z`, captured starting 2026-10-13T21:57:56.746726Z.

### 5.2 Second record

| **Parameter** | **Channel 0** | **Channel 1** |
|---|---|---|
| Start time | 2026-10-13 22:08:41 UTC | |
| Frequency | 1575.420 MHz (L1) | 1176.450 MHz (L5) |
| Duration | 300.0 s | |
| Sample rate | 2.000 MS/s | |
| Sample type | signed complex 16-bit samples (I+Q), little-endian | |
| Gain | 20.0 dB | 20.0 dB |
| Antenna | RX1 | RX2 |
| File size | 2400.0 MB | 2400.0 MB |
| Samples written | 600,000,000 | 600,000,000 |
| Lost samples | 0 | 0 |
| Clock source | external | |
| Time source | internal | |
| Motherboard | USRP X310 | |
| Daughterboard | TwinRX RX0 | TwinRX RX1 |
| Recorder | UHD Python API | |

Table 4: Metadata for recording `x310_20261013T220835Z`, captured starting 2026-10-13T22:08:41.589855Z.

# 6 Records on 22/10/2026

## 6.1 First record

| **Parameter** | **Channel 0** | **Channel 1** |
|---|---|---|
| Start time | 2026-10-22 19:00:06 UTC | |
| Frequency | 1575.420 MHz (L1) | 1176.450 MHz (L5) |
| Duration | 300.0 s | 300.0 s |
| Sample rate | 4.000 MS/s | 20.000 MS/s |
| Sample type | signed complex 8-bit samples (I+Q), converted on the host from the X310's sc16 | |
| Gain | 25.0 dB | 25.0 dB |
| Antenna | RX1 | RX2 |
| File size | 2400.0 MB | 12000.0 MB |
| Samples written | 1,200,000,000 | 6,000,000,000 |
| Lost samples | 2912 | 14924 |
| Clock source | external | |
| Time source | internal | |
| Motherboard | USRP X310 | |
| Daughterboard | TwinRX RX0 | TwinRX RX1 |
| Recorder | UHD Python API | |
| GNSS-SDR configuration | `l1ca8bits4msps_l5_8bits20msps.conf` | |

Table 5: Metadata for recording `x310_20261022T190000Z`, captured starting 2026-10-22T19:00:06.973177Z.

### 6.1.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

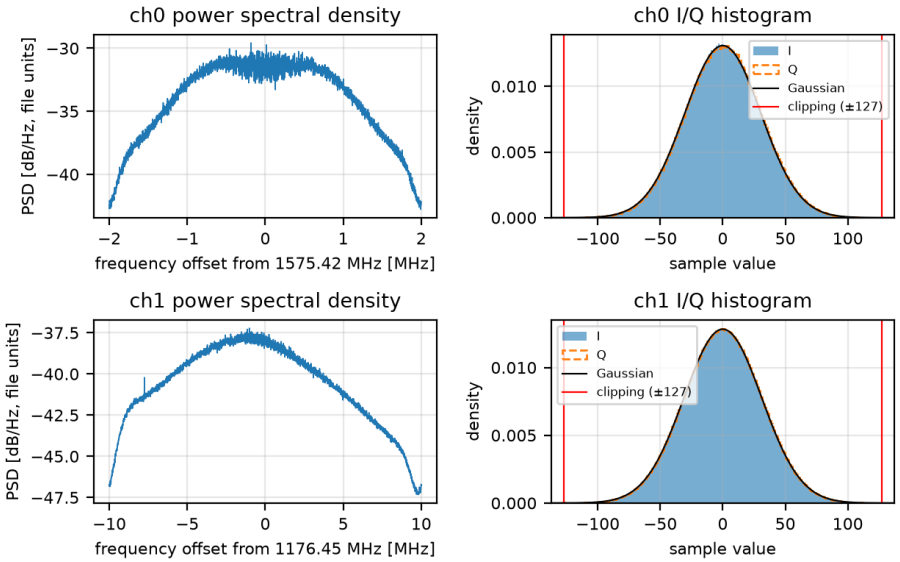



Figure 3: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and I/Q histograms (Gaussian noise, centred, not clipped).

| **Channel** | **ch0** | **ch1** |
|---|---|---|
| Frequency [MHz] | 1575.42 | 1176.45 |
| Rate [MS/s] | 4 | 20 |
| Format | ci8 | ci8 |
| Bytes per complex sample | 2 (1 for I, 1 for Q) | 2 (1 for I, 1 for Q) |
| Full scale | ±127 | ±127 |
| Gain [dB] | 25 | 25 |
| Std I / Q | 30.4 / 30.5 | 31.1 / 31.1 |
| Mean I / Q | -0.26 / 0.20 | 0.26 / -0.07 |
| Min / max I | -127 / 127 | -127 / 127 |
| Min / max Q | -127 / 127 | -127 / 127 |
| Headroom (full scale / std) | 4.2 | 4.1 |
| Clipped [%] | 0.003 | 0.004 |
| Lost samples | 2,912 | 14,924 |

Table 6: Signal source: sample statistics per channel (headroom: how many standard deviations of the noise fit before clipping).

### 6.1.2 Signal conditioner

| Channel | ch0 (L1 C/A) | ch1 (L5) |
|---|---|---|
| File format | ci8 | ci8 |
| Source item type | ibyte | ibyte |
| Data type adapter | Ibyte_To_Complex | Ibyte_To_Complex |
| Input filter | Pass_Through | Pass_Through |
| Resampling | ×5 zero-order hold, offline (`upsample_iq.py`) | Pass_Through |
| Rate [MS/s] | 4 → 20 | 20 → 20 |

Table 7: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

### 6.1.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

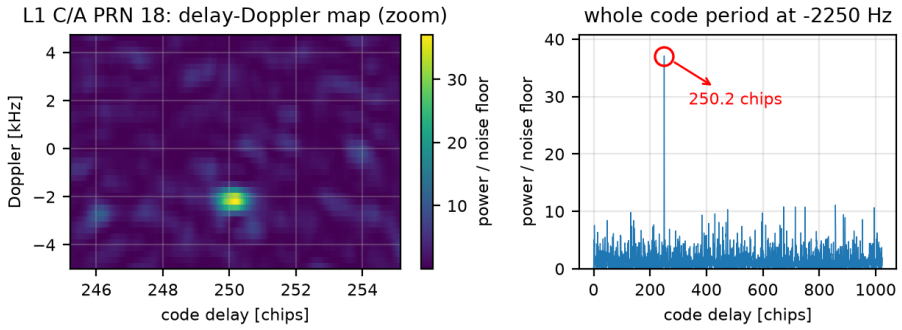



Figure 4: Acquisition, L1 C/A: delay–Doppler search of PRN 18; one sharp peak (test statistic 50 vs threshold 47) at -2250 Hz and 250.2 chips.

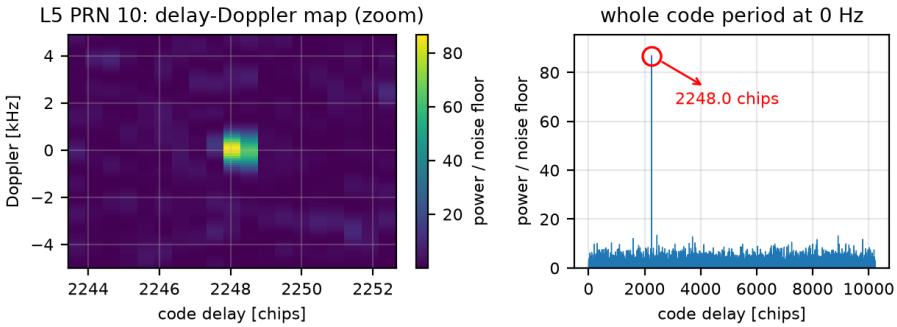



Figure 5: Acquisition, L5: delay–Doppler search of PRN 10; one sharp peak (test statistic 122 vs threshold 49) at -1491 Hz and 2248.0 chips.

| Band | Attempts | Detections | Detected PRNs | Best statistic / threshold |
|---|---|---|---|---|
| L1 C/A | 9 | 1 | 18 | 1.1 |
| L5 | 1 | 1 | 10 | 2.5 |

Table 8: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or, for L5/L2C, do not broadcast that signal) are rejected below the threshold.

### 6.1.4 Tracking

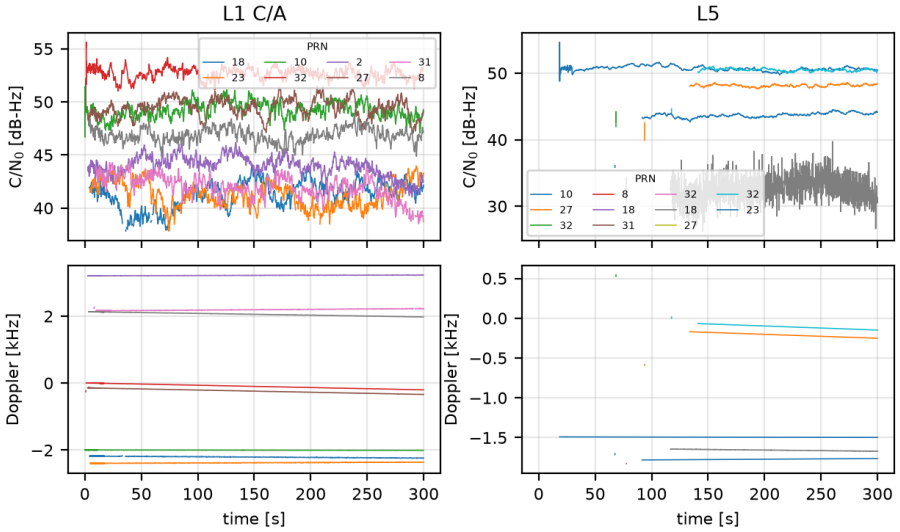



Figure 6: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

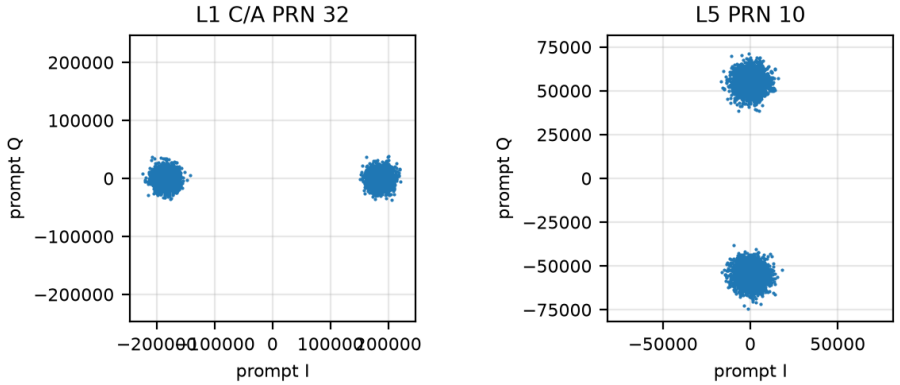



Figure 7: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis) mean the PLL is phase-locked; a ring would mean it is not. `Prompt_I`/`Prompt_Q` are 32-bit floats (sums of the samples over one integration period, in file units): they never clip, only overflowing beyond $\pm 3.4 \times 10^{38}$, far above the $N \cdot$ full scale $\cdot \sqrt{2}$ a sum of $N$ samples can reach.

| Band | Ch. | PRN | Median $C/N_0$ [dB-Hz] | Phase-locked [%] | Losses of lock | Tracked [s] |
|---|---|---|---|---|---|---|
| L1 C/A | 0 | 18 | 41.7 | 93 | 0 | 29 |
| L1 C/A | 1 | 23 | 41.9 | 93 | 0 | 29 |
| L1 C/A | 2 | 10 | 49.2 | 100 | 0 | 512 |
| L1 C/A | 3 | 32 | 52.9 | 100 | 0 | 30 |
| L1 C/A | 4 | 2 | 43.9 | 95 | 0 | 535 |
| L1 C/A | 5 | 27 | 49.1 | 94 | 0 | 532 |
| L1 C/A | 6 | 31 | 42.7 | 94 | 0 | 29 |
| L1 C/A | 7 | 8 | 47.0 | 94 | 0 | 30 |
| L5 | 8 | 10 | 50.7 | 100 | 0 | 381 |
| L5 | 10 | 27 | 48.0 | 37 | 2 | 45 |
| L5 | 10 | 32 | 29.0 | 0 | 2 | 16 |
| L5 | 11 | 8 | 28.4 | 0 | 2 | 16 |
| L5 | 11 | 18 | 33.5 | 0 | 2 | 16 |
| L5 | 12 | 31 | 29.0 | 0 | 2 | 16 |
| L5 | 12 | 32 | 28.7 | 0 | 2 | 16 |
| L5 | 13 | 18 | 32.8 | 60 | 1 | 306 |
| L5 | 13 | 27 | 28.8 | 0 | 1 | 16 |
| L5 | 14 | 32 | 50.3 | 36 | 1 | 44 |
| L5 | 15 | 23 | 43.7 | 63 | 0 | 332 |

Table 9: Tracking: per channel and satellite (phase-locked: carrier lock test $> 0.8$).

### 6.1.5 Telemetry decoder

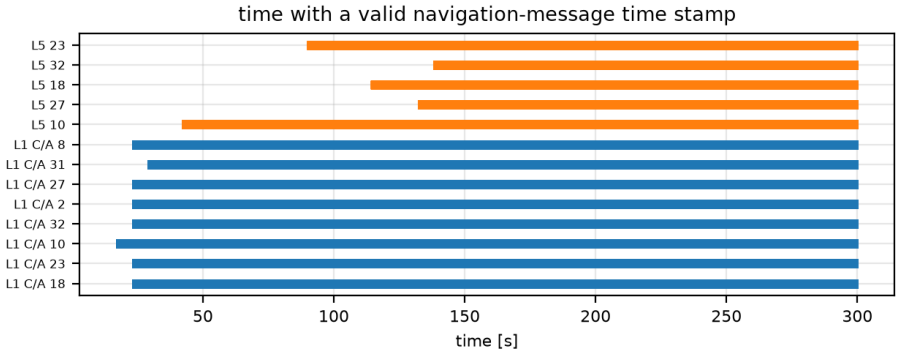



Figure 8: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

| Band | PRN | Messages decoded | Ephemeris | First valid at [s] | TOW [s] |
|---|---|---|---|---|---|
| L1 C/A | 2 | 47 | yes | 23.1 | 414048.0 |
| L1 C/A | 8 | 45 | yes | 23.1 | 414048.0 |
| L1 C/A | 10 | 43 | yes | 17.1 | 414042.0 |
| L1 C/A | 18 | 47 | yes | 23.1 | 414048.0 |
| L1 C/A | 23 | 47 | yes | 23.1 | 414048.0 |
| L1 C/A | 27 | 47 | yes | 23.1 | 414048.0 |
| L1 C/A | 31 | 46 | yes | 29.1 | 414054.0 |
| L1 C/A | 32 | 47 | yes | 23.1 | 414048.0 |
| L5 | 10 | 26 | yes | 42.2 | 414067.1 |
| L5 | 18 | 24 | yes | 114.4 | 414139.3 |
| L5 | 23 | 27 | yes | 90.0 | 414114.9 |
| L5 | 27 | 17 | yes | 132.3 | 414157.2 |
| L5 | 32 | 16 | yes | 138.2 | 414163.1 |

Table 10: Telemetry decoder: L1 C/A LNAV subframes and L2C/L5 CNAV messages decoded, ephemeris obtained, and the first valid time of week.

### 6.1.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

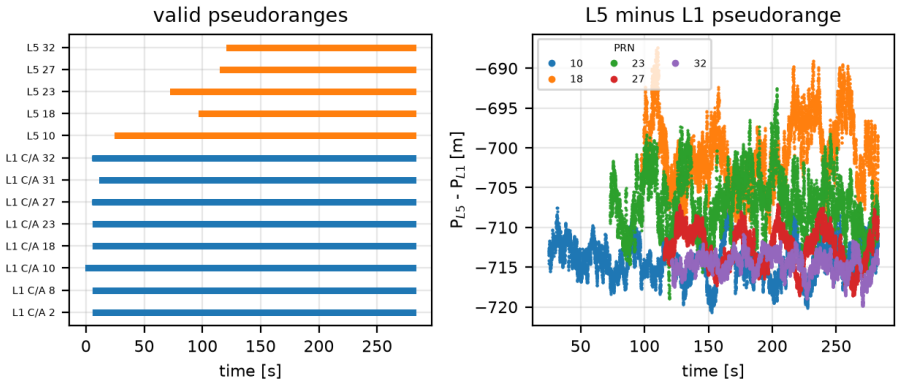



Figure 9: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

| PRN | Common epochs | of run [%] | Mean [m] | Std [m] | Drift [m/s] |
|---|---|---|---|---|---|
| 10 | 12831 | 86 | -714.11 | 2.26 | 0.0005 |
| 18 | 9267 | 62 | -700.79 | 4.40 | 0.0122 |
| 23 | 10483 | 70 | -706.92 | 3.69 | -0.0064 |
| 27 | 8373 | 56 | -712.59 | 2.40 | -0.0092 |
| 32 | 8078 | 54 | -714.63 | 1.36 | -0.0046 |

Table 11: Observables: $P_{L5} - P_{L1}$ per satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

### 6.1.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

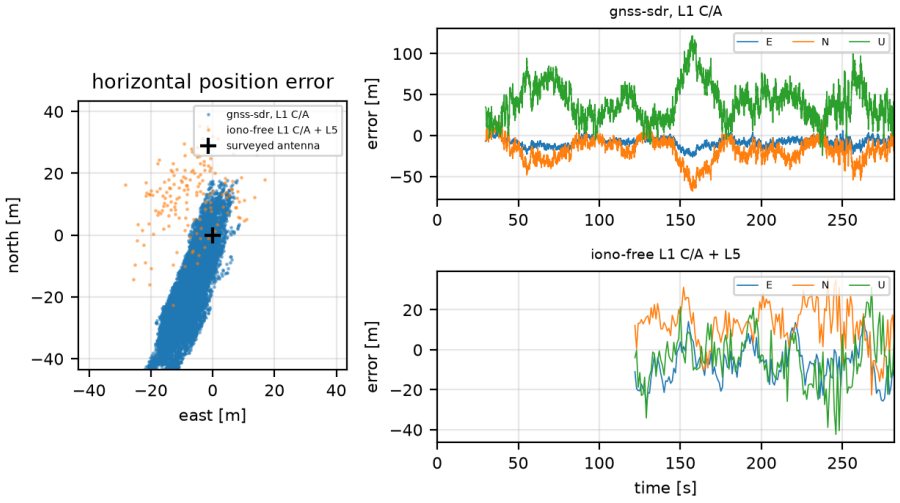



Figure 10: PVT: position error with respect to the surveyed antenna position. L1 C/A: gnss-sdr's own solution (broadcast ionosphere model). Iono-free: computed from gnss-sdr's observables and decoded ephemeris (gnss_spp.py), removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

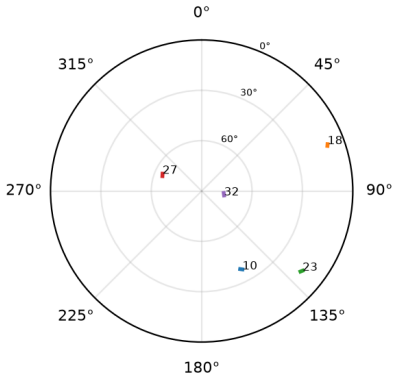



Figure 11: PVT: sky plot of the satellites in the dual-frequency solution (azimuth, elevation) during the recording.

| **Solution** | **gnss-sdr, L1 C/A** | **iono-free L1 C/A + L5** |
|---|---|---|
| Solutions | 13135 | 161 |
| Run with fix [%] | 90 | 57 |
| Mean E / N / U [m] | -7.6 / -18.0 / 40.8 | -7.3 / 12.2 / -4.6 |
| Std E / N / U [m] | 5.5 / 14.2 / 22.4 | 9.3 / 10.4 / 12.8 |
| RMS 2-D [m] | 24.8 | 19.9 |
| RMS 3-D [m] | 52.8 | 24.1 |
| 95% horiz. [m] | 46.5 | 29.6 |

Table 12: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, the iono-free one is computed once per second).